

NICHOLAS SKIBA

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SUMMARY

Mechanical Engineering student in a Co-Terminal B.S./M.S. program (3.59 GPA) with experience in mechanical design, RF/EMI systems, and CAD modeling. Strong foundation in Python, MATLAB, and FEA tools, with teamwork and leadership from athletics, business, and internship experience.

EDUCATION

Illinois Institute of Technology - Chicago, IL August 2023 - December 2027 (Expected)

Co-Terminal Program: B.S. Mechanical Engineering / M.S. Mechanical and Aerospace Engineering

Minor: Engineering Graphics and CAD | GPA: 3.59

Coursework: Thermodynamics, Fluid Mechanics, Advanced Mechanics of Solids, Advanced CAD/CAM/CAE, Manufacturing Processes, Design of Machine Elements, Mass and Heat Transfer, Dynamics, Computational Mechanics, Materials Science, Mechanical Lab, Design for Innovation, Engineering Drawing and Design

SKILLS

Programming: Python, MATLAB, G-Code, Arduino, LaTeX, Mathematica

Design/Simulation: Inventor, AutoCAD, SolidWorks, Mastercam, Revit, FEA (SolidWorks, Inventor), GD&T, ANSYS

Fabrication/Tools: 3D Printing, Welding, Plasma Cutting, Soldering, Laser Cutting

Instruments: Vector Network Analyzer (VNA), Oscilloscopes, RF Amplifiers, Signal Generators

EXPERIENCE

Engineering Intern - Deep Coat Industries - Sugar Grove, IL May 2025 - Aug 2025; Dec 2025 - Jan 2026

- Designed, built, and validated a full-spectrum EMI/RFI test system covering 500 Hz to 6.3 GHz, including an anechoic chamber and cassette-style sample holder for rapid, repeatable tests
- Engineered and fabricated transmitting and receiving antennas (loop, monopole, etc.) and custom fixtures (3D-printed and machined) to ensure precise alignment and measurement repeatability
- Automated VNA and oscilloscope control, data acquisition, and analysis with Python, generating visualizations, trendlines, and predictive models for high-frequency performance
- Led mechanical design of a compact 90-degree rotary mechanism: full AutoCAD modeling, detailed part/assembly drawings, and manufacturing/assembly documentation
- Performed hands-on fabrication (plasma cutting, etc.) and documented procedures for handoff

Business Owner/Operator - Triple Threat Detailing - IL May 2024 - Present

- Founded and operating a car-detailing business; managing marketing, finance, scheduling, and customer relations while delivering high-quality service, maintaining five-star client satisfaction

Sales Associate - Runnings - Homer Glen, IL June 2021 - May 2025

PROJECTS

Four-Bar EV Charging Arm ([GitHub](#)) November 2025

- Developed automatic EV charging arm using four-bar linkage; optimized geometry in MATLAB (fmincon) reducing torque by 70%, and Arduino control with IR sensor input (<2 mm accuracy)

Sustainable Foam-Core Chair ([GitHub](#)) September 2025

- Devised and tested foam-core chair supporting 103 kg; optimized structure using Inventor FEA

Bio-Inspired Hexapod Robot ([GitHub](#)) December 2025

- Designed a six-legged tripod-gait robot using Arduino and servo actuation; integrated mechanical and electrical systems and validated 10-meter traversal at 1.3 body lengths/s (0.24 m/s)

ACTIVITIES, LEADERSHIP, AND HONORS

Illinois Tech Men's Soccer August 2023 - Present

- Balancing academics and athletics (20+ hr/week), time management, and organizational skills

Dean's List - Illinois Institute of Technology August 2023 - December 2025